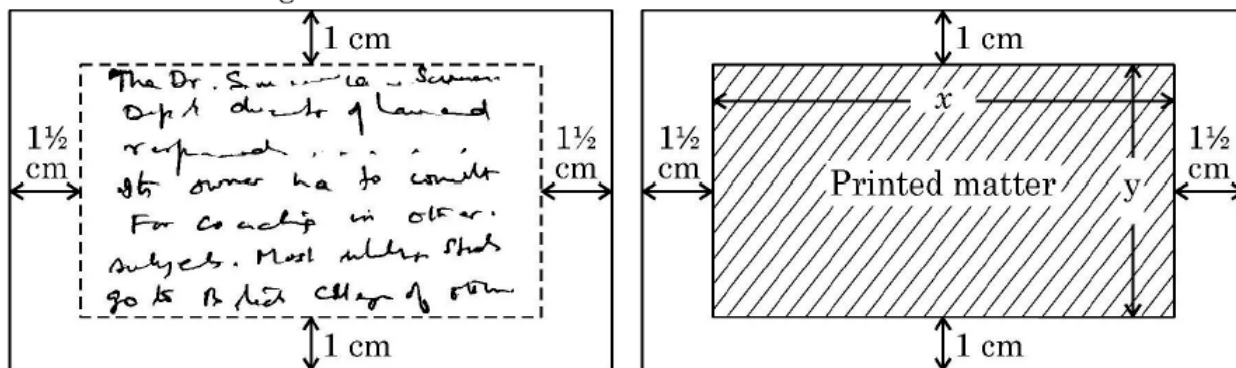


UNIT-3 CALCULUS

CASE BASED QUESTIONS

3.

A rectangular visiting card is to contain 24 sq.cm. of printed matter. The margins at the top and bottom of the card are to be 1 cm and the margins on the left and right are to be $1\frac{1}{2}$ cm as shown below :



On the basis of the above information, answer the following questions :

- (i) Write the expression for the area of the visiting card in terms of x .
- (ii) Obtain the dimensions of the card of minimum area.

4.

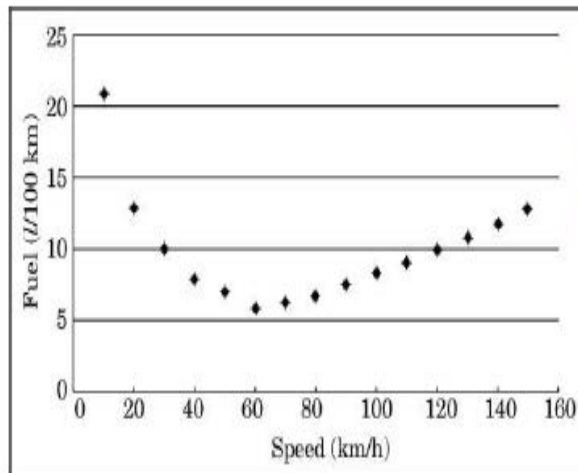
A store has been selling calculators at ₹ 350 each. A market survey indicates that a reduction in price (p) of calculator increases the number of units (x) sold. The relation between the price and quantity sold is given by the demand function $p = 450 - \frac{1}{2}x$.



Based on the above information, answer the following questions :

- (i) Determine the number of units (x) that should be sold to maximise the revenue $R(x) = xp(x)$. Also, verify the result.
- (ii) What rebate in price of calculator should the store give to maximise the revenue ?

5. Overspeeding increases fuel consumption and decreases fuel economy as a result of tyre rolling friction and air resistance. While vehicles reach optimal fuel economy at different speeds, fuel mileage usually decreases rapidly at speeds above 80 km/h.



The relation between fuel consumption F (l/100 km) and speed V (km/h) under some constraints is given as $F = \frac{V^2}{500} - \frac{V}{4} + 14$.

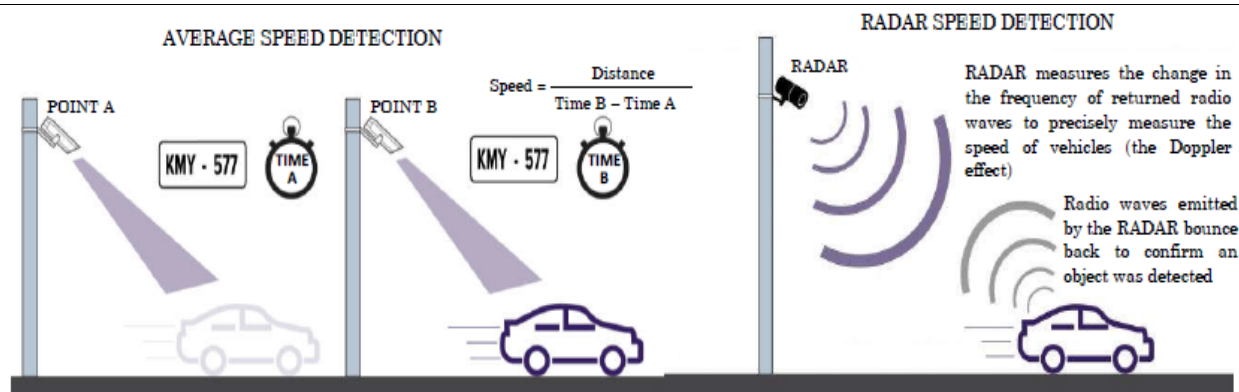
On the basis of the above information, answer the following questions :

- (i) Find F , when $V = 40$ km/h.
- (ii) Find $\frac{dF}{dV}$.
- (iii) (a) Find the speed V for which fuel consumption F is minimum.

OR

- (iii) (b) Find the quantity of fuel required to travel 600 km at the speed V at which $\frac{dF}{dV} = -0.01$.

6. The traffic police has installed Over Speed Violation Detection (OSVD) system at various locations in a city. These cameras can capture a speeding vehicle from a distance of 300 m and even function in the dark.



A camera is installed on a pole at the height of 5 m. It detects a car travelling away from the pole at the speed of 20 m/s. At any point, x m away from the base of the pole, the angle of elevation of the speed camera from the car C is θ .

On the basis of the above information, answer the following questions :

- (i) Express θ in terms of height of the camera installed on the pole and x .
- (ii) Find $\frac{d\theta}{dx}$.
- (iii) (a) Find the rate of change of angle of elevation with respect to time at an instant when the car is 50 m away from the pole.

OR

- (iii) (b) If the rate of change of angle of elevation with respect to time of another car at a distance of 50 m from the base of the pole is $\frac{3}{101}$ rad/s, then find the speed of the car.

7.

A tank, as shown in the figure below, formed using a combination of a cylinder and a cone, offers better drainage as compared to a flat bottomed tank.



A tap is connected to such a tank whose conical part is full of water. Water is dripping out from a tap at the bottom at the uniform rate of $2 \text{ cm}^3/\text{s}$. The semi-vertical angle of the conical tank is 45° .

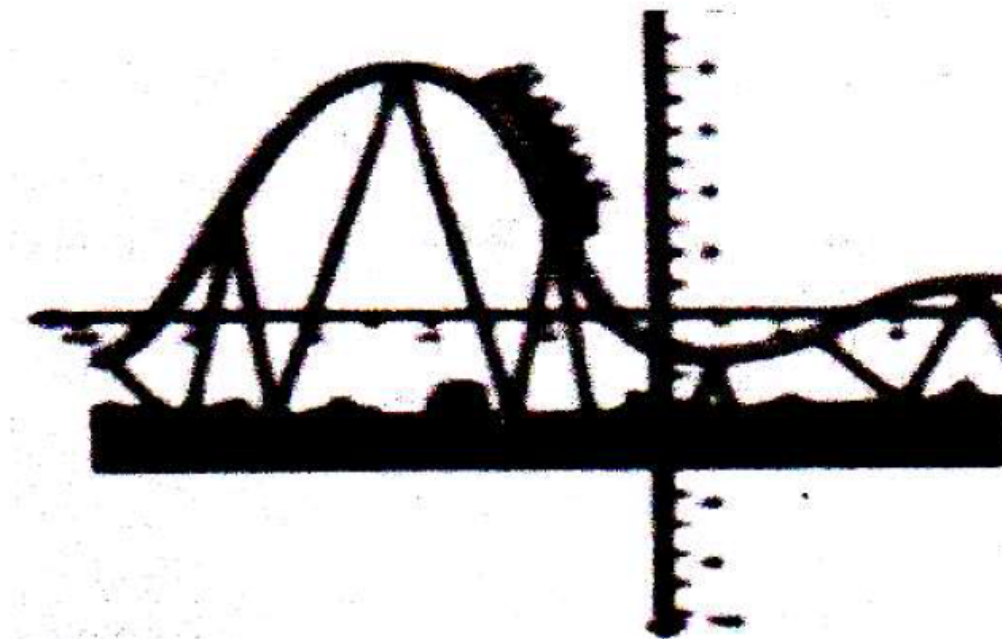
On the basis of given information, answer the following questions :

- (i) Find the volume of water in the tank in terms of its radius r .
- (ii) Find rate of change of radius at an instant when $r = 2\sqrt{2} \text{ cm}$.
- (iii) (a) Find the rate at which the wet surface of the conical tank is decreasing at an instant when radius $r = 2\sqrt{2} \text{ cm}$.

OR

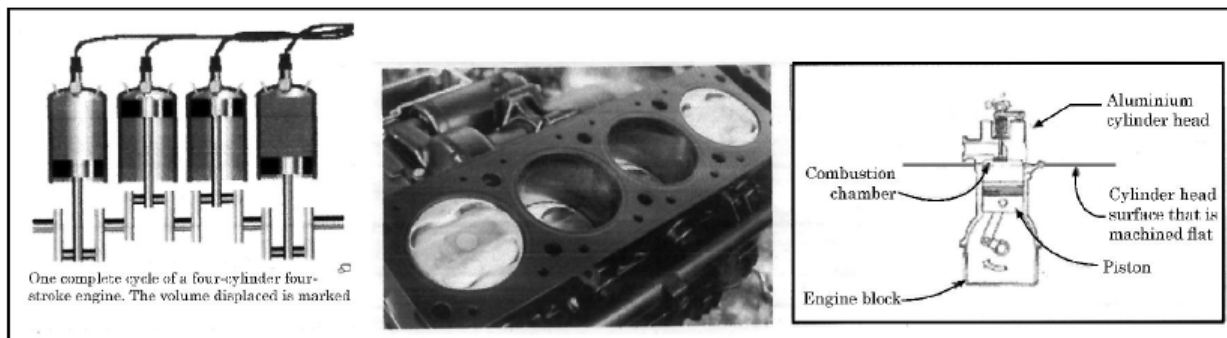
- (iii) (b) Find the rate of change of height 'h' at an instant when slant height is 4 cm.

The equation of the path traced by a roller-coaster is given by the polynomial $f(x) = a(x + 9)(x + 1)(x - 3)$. If the roller-coaster crosses y-axis at a point $(0, -1)$, answer the following :



- (i) Find the value of 'a'.
- (ii) Find $f''(x)$ at $x = 1$.

8. Engine displacement is the measure of the cylinder volume swept by all the pistons of a piston engine. The piston moves inside the cylinder bore



The cylinder bore in the form of circular cylinder open at the top is to be made from a metal sheet of area $75\pi \text{ cm}^2$.

Based on the above information, answer the following questions :

- (i) If the radius of cylinder is $r \text{ cm}$ and height is $h \text{ cm}$, then write the volume V of cylinder in terms of radius r .
- (ii) Find $\frac{dV}{dr}$.
- (iii) (a) Find the radius of cylinder when its volume is maximum.

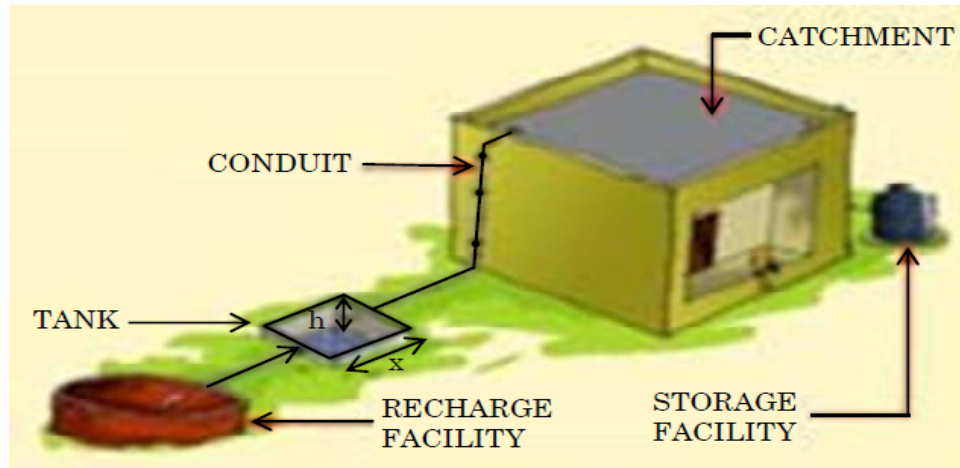
OR

- (b) For maximum volume, $h > r$. State true or false and justify.

9.

In order to set up a rain water harvesting system, a tank to collect rain water is to be dug. The tank should have a square base and a capacity of 250 m^3 . The cost of land is ₹ 5,000 per square metre and cost of digging increases with depth and for the whole tank, it is ₹ $40,000 h^2$, where h is the depth of the tank in metres. x is the side of the square base of the tank in metres.

ELEMENTS OF A TYPICAL RAIN WATER HARVESTING SYSTEM



- (iii) (a) Find the value of x for which cost C is minimum.

OR

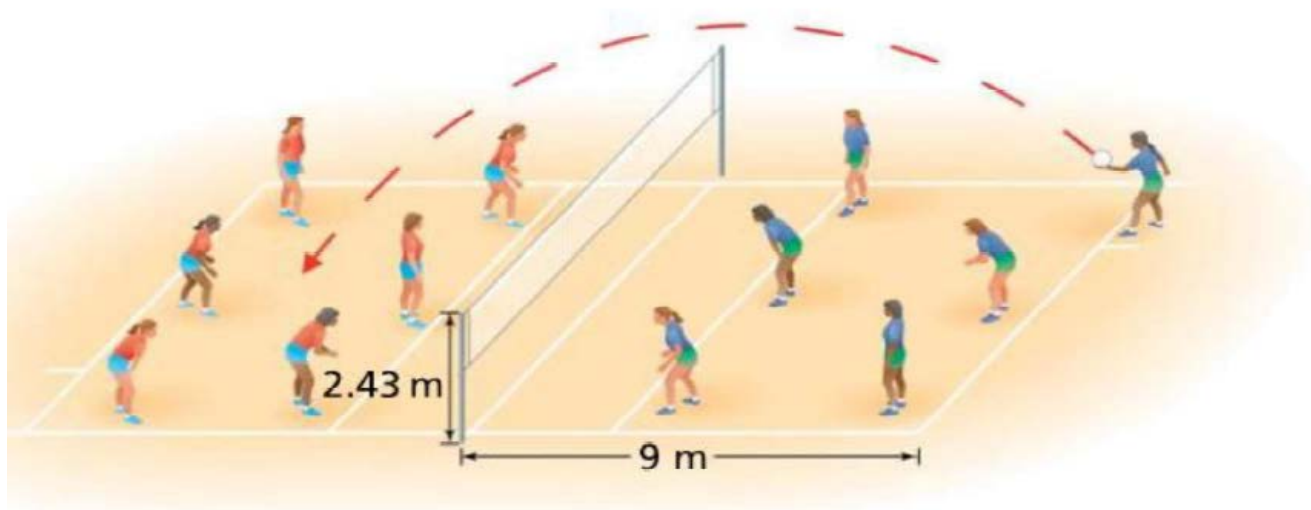
- (iii) (b) Check whether the cost function $C(x)$ expressed in terms of x is increasing or not, where $x > 0$.

Based on the above information, answer the following questions :

- (i) Find the total cost C of digging the tank in terms of x .
 (ii) Find $\frac{dC}{dx}$.

10.

A volleyball player serves the ball which takes a parabolic path given by the equation $h(t) = -\frac{7}{2}t^2 + \frac{13}{2}t + 1$, where $h(t)$ is the height of ball at any time t (in seconds), ($t \geq 0$).

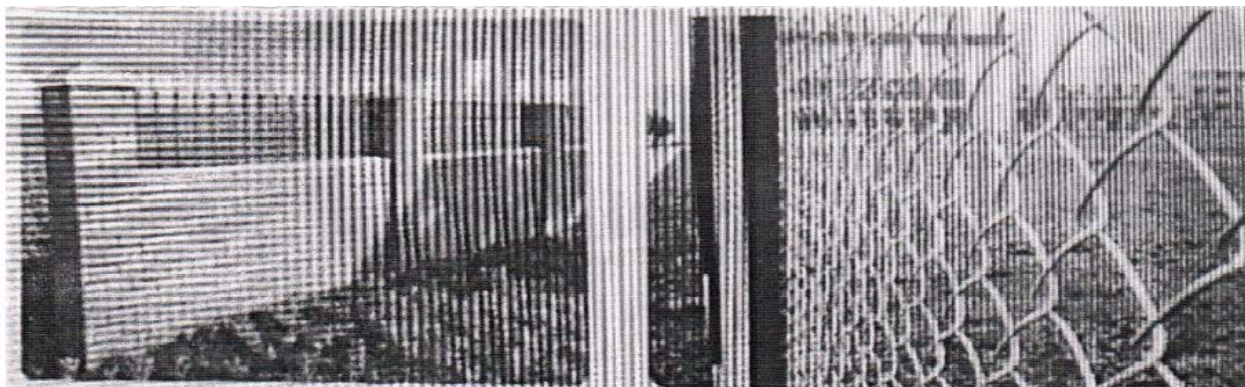


Based on the above information, answer the following questions :

- (i) Is $h(t)$ a continuous function ? Justify.
- (ii) Find the time at which the height of the ball is maximum.

11.

Sooraj's father wants to construct a rectangular garden using a brick wall on one side of the garden and wire fencing for the other three sides as shown in the figure. He has 200 metres of fencing wire.

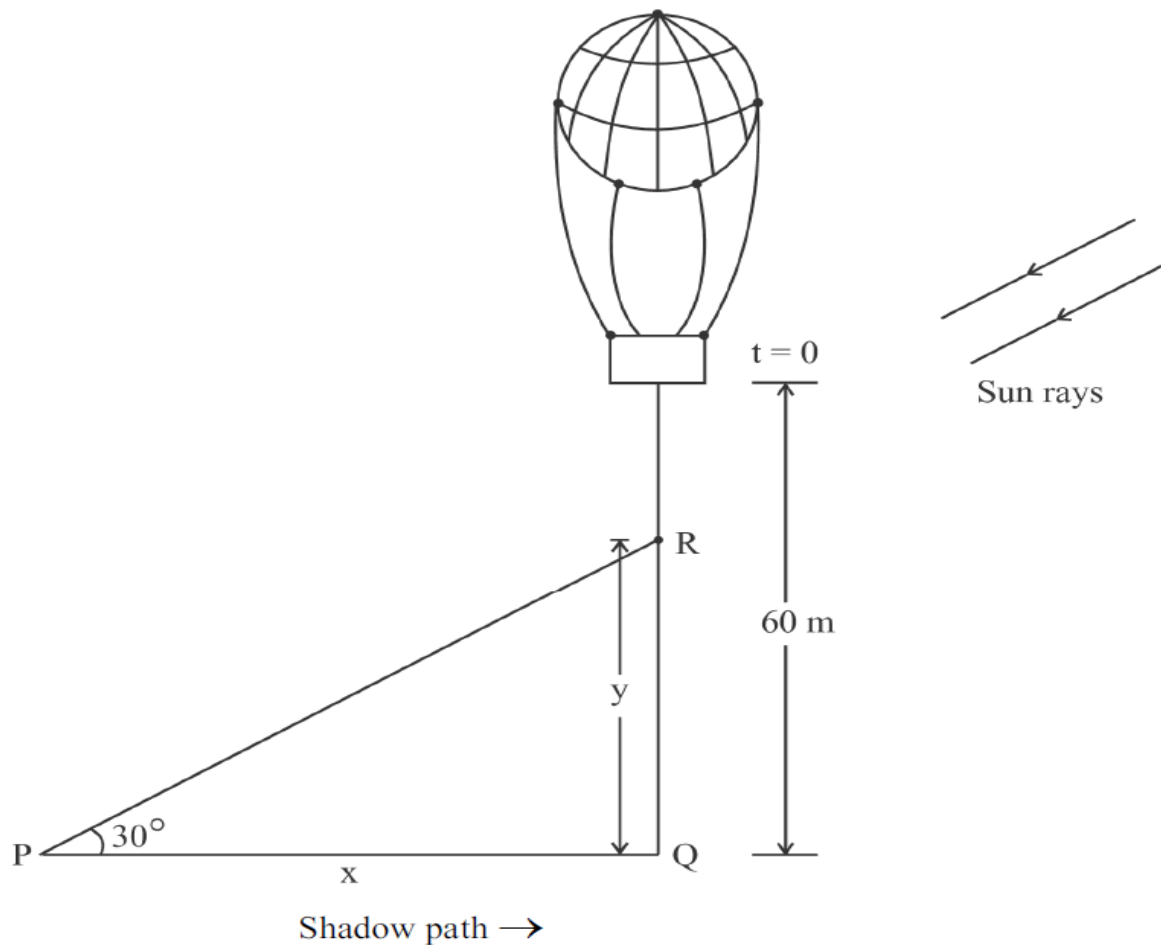


Based on the above information, answer the following questions :

- (i) Let ' x ' metres denote the length of the side of the garden perpendicular to the brick wall and ' y ' metres denote the length of the side parallel to the brick wall. Determine the relation representing the total length of fencing wire and also write $A(x)$, the area of the garden.
- (ii) Determine the maximum value of $A(x)$.

12.

A sandbag is dropped from a balloon at a height of 60 metres.



When the angle of elevation of the sun is 30° , the position of the sandbag is given by the equation $y = 60 - 4.9 t^2$, where y is the height of the sandbag above the ground and t is the time in seconds.

On the basis of the above information, answer the following questions :

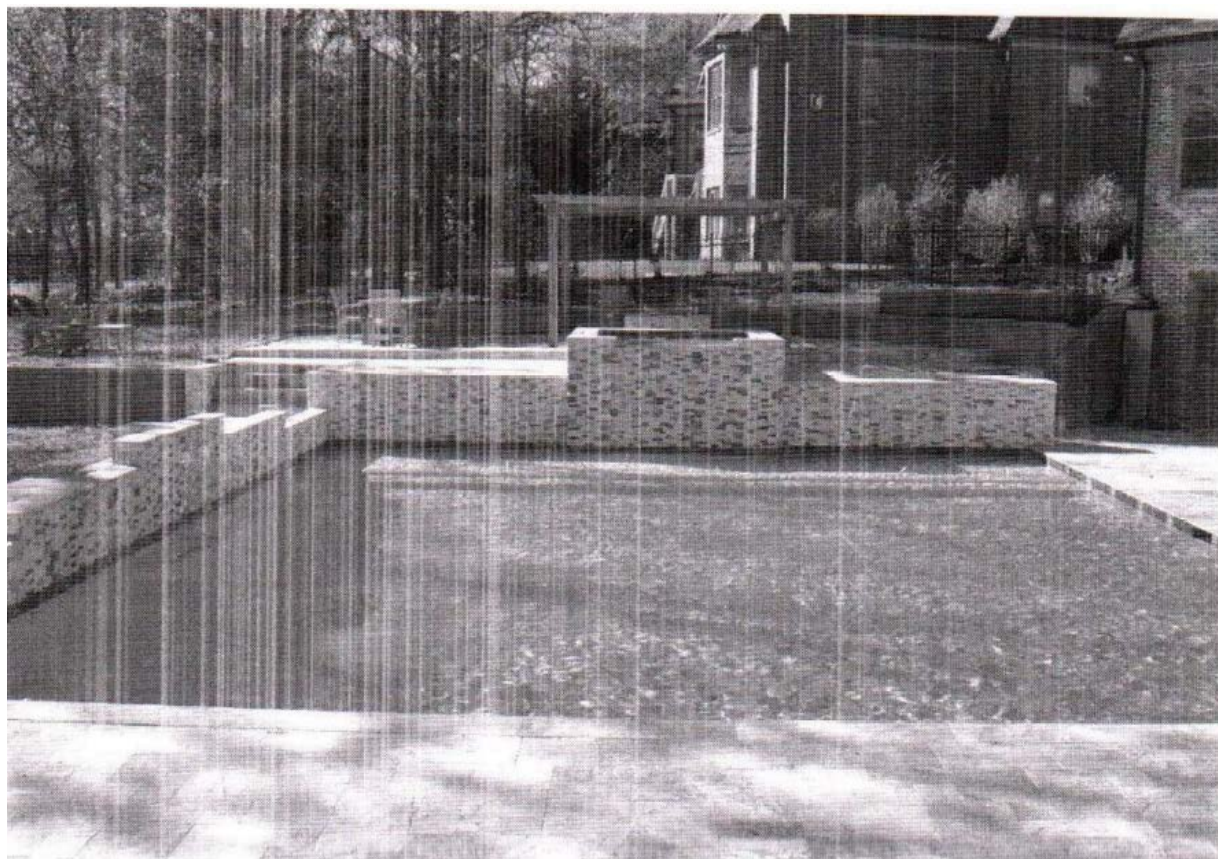
- (i) Find the relation between x and y , where x is the distance of the shadow at P from the point Q and y is the height of the sandbag above the ground.
- (ii) After how much time will the sandbag be 35 metres above the ground ?
- (iii) (a) Find the rate at which the shadow of the sandbag is travelling along the ground when the sandbag is at a height of 35 metres.

OR

- (iii) (b) How fast is the height of the sandbag decreasing when 2 seconds have elapsed ?

13.

A housing society wants to commission a swimming pool for its residents. For this, they have to purchase a square piece of land and dig this to such a depth that its capacity is 250 cubic metres. Cost of land is ₹ 500 per square metre. The cost of digging increases with the depth and cost for the whole pool is ₹ $4000(\text{depth})^2$.



Suppose the side of the square plot is x metres and depth is h metres.

On the basis of the above information, answer the following questions :

- (i) Write cost $C(h)$ as a function in terms of h .
- (ii) Find critical point.
- (iii) (a) Use second derivative test to find the value of h for which cost of constructing the pool is minimum. What is the minimum cost of construction of the pool ?

OR

- (iii) (b) Use first derivative test to find the depth of the pool so that cost of construction is minimum. Also, find relation between x and h for minimum cost.

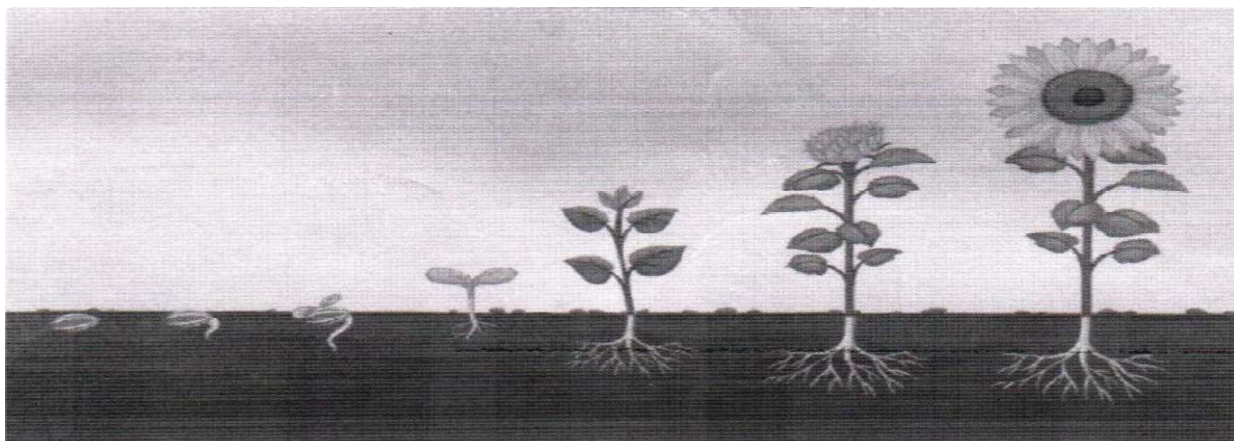
14.

In an agricultural institute, scientists do experiments with varieties of seeds to grow them in different environments to produce healthy plants and get more yield.

A scientist observed that a particular seed grew very fast after germination. He had recorded growth of plant since germination and he said that its growth can be defined by the function

$$f(x) = \frac{1}{3}x^3 - 4x^2 + 15x + 2, \quad 0 \leq x \leq 10$$

where x is the number of days the plant is exposed to sunlight.



On the basis of the above information, answer the following questions :

- (i) What are the critical points of the function $f(x)$?
- (ii) Using second derivative test, find the minimum value of the function.

15.

Let $f(x)$ be a real valued function. Then its

- Left Hand Derivative (L.H.D.) : $Lf'(a) = \lim_{h \rightarrow 0} \frac{f(a-h) - f(a)}{-h}$
- Right Hand Derivative (R.H.D.) : $Rf'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$

Also, a function $f(x)$ is said to be differentiable at $x = a$ if its L.H.D. and R.H.D. at $x = a$ exist and both are equal.

For the function $f(x) = \begin{cases} |x-3|, & x \geq 1 \\ \frac{x^2}{4} - \frac{3x}{2} + \frac{13}{4}, & x < 1 \end{cases}$

answer the following questions :

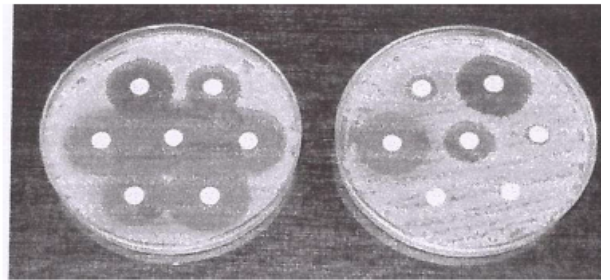
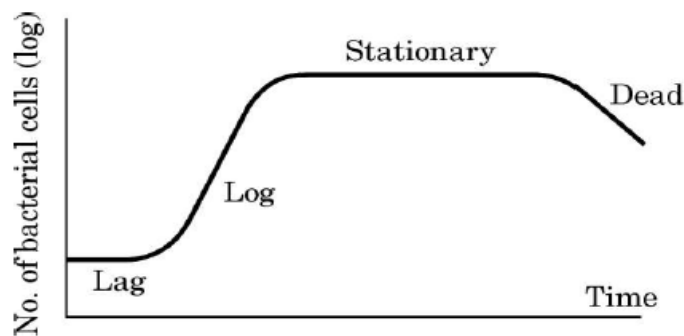
- (i) What is R.H.D. of $f(x)$ at $x = 1$?
- (ii) What is L.H.D. of $f(x)$ at $x = 1$?
- (iii) (a) Check if the function $f(x)$ is differentiable at $x = 1$.

OR

- (iii) (b) Find $f'(2)$ and $f'(-1)$.

16.

A bacteria sample of certain number of bacteria is observed to grow exponentially in a given amount of time. Using exponential growth model, the rate of growth of this sample of bacteria is calculated.



The differential equation representing the growth of bacteria is given as :

$$\frac{dP}{dt} = kP, \text{ where } P \text{ is the population of bacteria at any time 't'.$$

Based on the above information, answer the following questions :

- (i) Obtain the general solution of the given differential equation and express it as an exponential function of 't'.
- (ii) If population of bacteria is 1000 at $t = 0$, and 2000 at $t = 1$, find the value of k .

17.

An equation involving derivatives of the dependent variable with respect to the independent variables is called a differential equation. A differential equation of the form $\frac{dy}{dx} = F(x, y)$ is said to be homogeneous if

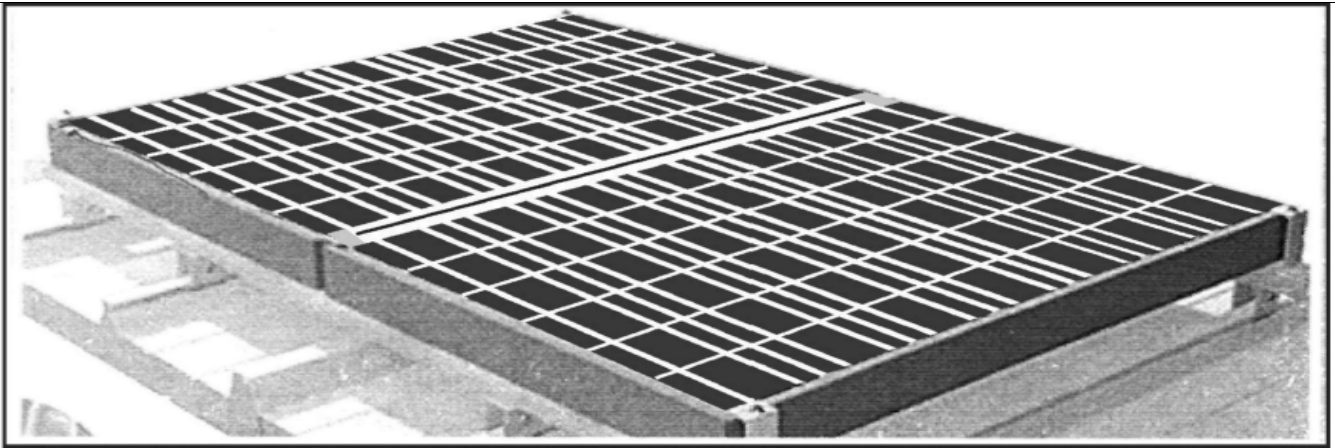
$F(x, y)$ is a homogeneous function of degree zero, whereas a function $F(x, y)$ is a homogenous function of degree n if $F(\lambda x, \lambda y) = \lambda^n F(x, y)$. To solve a homogeneous differential equation of the type $\frac{dy}{dx} = F(x, y) =$

$g\left(\frac{y}{x}\right)$, we make the substitution $y = vx$ and then separate the variables.

Based on the above, answer the following questions :

- (I) Show that $(x^2 - y^2) dx + 2xy dy = 0$ is a differential equation of the type $\frac{dy}{dx} = g\left(\frac{y}{x}\right)$.
- (II) Solve the above equation to find its general solution.

18.



A technical company is designing a rectangular solar panel installation on a roof using 300 metres of boundary material. The design includes a partition running parallel to one of the sides dividing the area (roof) into two sections.

Let the length of the side perpendicular to the partition be x metres and with parallel to the partition be y metres.

Based on this information, answer the following questions :

- (i) Write the equation for the total boundary material used in the boundary and parallel to the partition in terms of x and y . 1
- (ii) Write the area of the solar panel as a function of x . 1
- (iii) (a) Find the critical points of the area function. Use second derivative test to determine critical points at the maximum area. Also, find the maximum area. 2

OR

- (iii) (b) Using first derivative test, calculate the maximum area the company can enclose with the 300 metres of boundary material, considering the parallel partition. 2

19.



A small town is analyzing the pattern of a new street light installation. The lights are set up in such a way that the intensity of light at any point x metres from the start of the street can be modelled by $f(x) = e^x \sin x$, where x is in metres.

Based on the above, answer the following :

- (i) Find the intervals on which the $f(x)$ is increasing or decreasing, $x \in [0, \pi]$. 2
- (ii) Verify, whether each critical point when $x \in [0, \pi]$ is a point of local maximum or local minimum or a point of inflexion. 2

- 20.** During a heavy gaming session, the temperature of a student's laptop processor increases significantly. After the session, the processor begins to cool down, and the rate of cooling is proportional to the difference between the processor's temperature and the room temperature (25°C). Initially the processor's temperature is 85°C . The rate of cooling is defined by the equation $\frac{d}{dt}(T(t)) = -k(T(t) - 25)$,

where $T(t)$ represents the temperature of the processor at time t (in minutes) and k is a constant.



Based on the above information, answer the following questions :

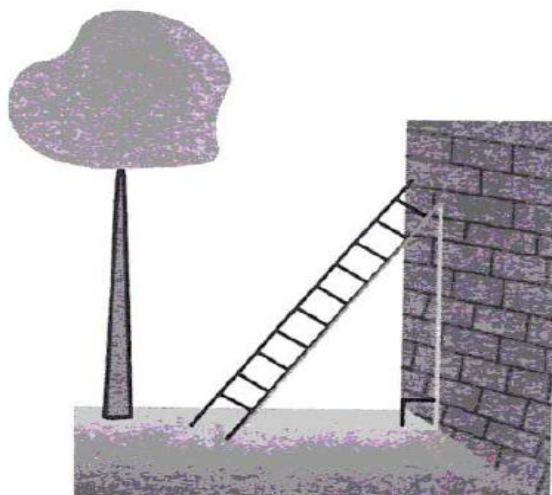
- (i) Find the expression for temperature of processor, $T(t)$ given that $T(0) = 85^\circ\text{C}$. 2
- (ii) How long will it take for the processor's temperature to reach 40°C ? Given that $k = 0.03$, $\log_e 4 = 1.3863$. 2

- 21.** A carpenter needs to make a wooden cuboidal box, closed from all sides, which has a square base and fixed volume. Since he is short of the paint required to paint the box on completion, he wants the surface area to be minimum.

On the basis of the above information, answer the following questions :

- (i) Taking length = breadth = x m and height = y m, express the surface area (S) of the box in terms of x and its volume (V), which is constant. 1
- (ii) Find $\frac{dS}{dx}$. 1
- (iii) (a) Find a relation between x and y such that the surface area (S) is minimum. 2
- OR**
- (iii) (b) If surface area (S) is constant, the volume (V) = $\frac{1}{4}(Sx - 2x^3)$, x being the edge of base. Show that volume (V) is maximum for $x = \sqrt{\frac{S}{6}}$. 2

22.

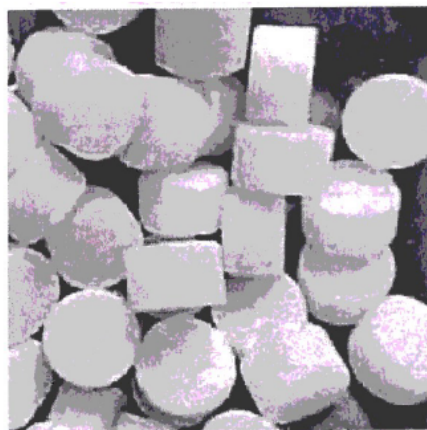


A ladder of fixed length ' h ' is to be placed along the wall such that it is free to move along the height of the wall.

Based upon the above information, answer the following questions :

- | | | |
|-----------|--|---|
| | (i) Express the distance (y) between the wall and foot of the ladder in terms of 'h' and height (x) on the wall at a certain instant. Also, write an expression in terms of h and x for the area (A) of the right triangle, as seen from the side by an observer. | 1 |
| | (ii) Find the derivative of the area (A) with respect to the height on the wall (x), and find its critical point. | 1 |
| | (iii) (a) Show that the area (A) of the right triangle is maximum at the critical point. | 2 |
| OR | | |
| | (iii) (b) If the foot of the ladder whose length is 5 m, is being pulled towards the wall such that the rate of decrease of distance (y) is 2 m/s, then at what rate is the height on the wall (x) increasing, when the foot of the ladder is 3 m away from the wall ? | 2 |

- 23.** Camphor is a waxy, colourless solid with strong aroma that evaporates through the process of sublimation, if left in the open at room temperature.



(Cylindrical-shaped Camphor tablets)

A cylindrical camphor tablet whose height is equal to its radius (r) evaporates when exposed to air such that the rate of reduction of its volume is proportional to its total surface area. Thus, $\frac{dV}{dt} = kS$ is the differential equation, where V is the volume, S is the surface area and t is the time in hours.

Based upon the above information, answer the following questions :

- (i) Write the order and degree of the given differential equation. 1
- (ii) Substituting $V = \pi r^3$ and $S = 2\pi r^2$, we get the differential equation $\frac{dr}{dt} = \frac{2}{3}k$. Solve it, given that $r(0) = 5$ mm. 1

- (iii) (a) If it is given that $r = 3$ mm when $t = 1$ hour, find the value of k . Hence, find t for $r = 0$ mm. 2

OR

- (iii) (b) If it is given that $r = 1$ mm when $t = 1$ hour, find the value of k . Hence, find t for $r = 0$ mm. 2

24. Read the following passage and answer the questions given below.



The temperature of a person during an intestinal illness is given by $f(x) = -0.1x^2 + mx + 98.6, 0 \leq x \leq 12$, m being a constant, where $f(x)$ is the temperature in $^{\circ}\text{F}$ at x days.

- (i) Is the function differentiable in the interval $(0, 12)$? Justify your answer.
- (ii) If 6 is the critical point of the function, then find the value of the constant m .
- (iii) Find the intervals in which the function is strictly increasing/strictly decreasing.
- OR
- (iii) Find the points of local maximum/local minimum, if any, in the interval $(0, 12)$ as well as the points of absolute maximum/absolute minimum in the interval $[0, 12]$. Also, find the corresponding local maximum/local minimum and the absolute maximum/absolute minimum values of the function.

25.



Read the following passage and answer the questions given below.

In an elliptical sport field the authority wants to design a rectangular soccer field with the maximum possible area. The sport field is given by the graph of

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

- (i) If the length and the breadth of the rectangular field be $2x$ and $2y$ respectively, then find the area function in terms of x .
- (ii) Find the critical point of the function.
- (iii) Use First derivative Test to find the length $2x$ and width $2y$ of the soccer field (in terms of a and b) that maximize its area.

OR

- (iii) Use Second Derivative Test to find the length $2x$ and width $2y$ of the soccer field (in terms of a and b) that maximize its area.

26.

Read the following passage and answer the questions given below:

The relation between the height of the plant (' y ' in cm) with respect to its exposure to the sunlight

is governed by the following equation $y = 4x - \frac{1}{2}x^2$, where ' x ' is the number of days exposed to the sunlight, for $x \leq 3$.



- (i) Find the rate of growth of the plant with respect to the number of days exposed to the sunlight.

	(ii) Does the rate of growth of the plant increase or decrease in the first three days? What will be the height of the plant after 2 days?
27.	Ramesh, the owner of a sweet selling shop, purchased some rectangular card board sheets of dimension 25cm by 40cm to make container packets without top. Let $x\text{cm}$ be the length of the side of the square to be cut out from each corner to give that sheet the shape of the container by folding up the flaps. Based on the above information answer the following questions. (i) Express the volume (V) of each container as function of x only. [1 Mark] (ii) Find $\frac{dV}{dx}$ [1 Mark] (iii) (a) For what value of x, the volume of each container is maximum? [2 Marks] OR (iii) (b) Check whether V has a point of inflection at $x = \frac{65}{6}$ or not? [2 Marks]
28.	LED bulbs are energy-efficient because they use significantly less electricity than traditional bulbs while producing the same amount of light. They convert more energy into light rather than heat, reducing waste. Additionally, their long lifespan means fewer replacements, saving resources and money over time. A company manufactures a new type of energy-efficient LED bulb. The cost of production and the revenue generated by selling x bulbs (in an hour) are modelled as $C(x) = 0.5x^2 - 10x + 150$ and $R(x) = -0.3x^2 + 20x$ respectively, where $C(x)$ and $R(x)$ are both in ₹. 

To maximize the profit, the company needs to analyze these functions using calculus. Use the given models to answer the following questions:

- I. Derive the profit function $P(x)$ [1]
- II. Find the critical points of $P(x)$. [1]
- III A. Determine whether the critical points correspond to a maximum or a minimum profit by using the second derivative test.

OR

- III B. Identify the possible practical value of x (i.e., the number of bulbs that can realistically be produced and sold) that can maximize the profit, if the resources available and the expenditure on machines allows to produce minimum 10 but not more than 18 bulbs per hour. Also calculate the maximum profit. [2]